

13-May-2026

Agenda:

FE-8

→ feature selection $x[i] \rightarrow y[\text{target}]$

- BE → p-value
- FS → p-value & R^2
- RFE → linear regression
- Random Forest

} correlation with targeted column

→ multicollinearity → within $x[i]$

→ vif

↓

variance inflation factor

$$vif = \frac{1}{1 - R^2} > 10, \text{ drop that variable}$$

$$\underset{vif}{10} = \frac{1}{1 - R^2} \Rightarrow R^2 \rightarrow 0.9 \xrightarrow{90\% \text{ of variance is explained}}$$

$$\xrightarrow{90\%} x[i] \rightarrow x[j]$$

$R^2 \rightarrow 1 \rightarrow 100\%$ variance is explained

$$vif = \frac{1}{1 - 1} = \infty$$

↓
redundant

$R^2 \rightarrow 0 \rightarrow 0\%$ variance is explained

$$vif = \frac{1}{1 - 0} = 1$$

↓
new information

lot Area, Year Built, CrLiv Area

X_i

Sale Price

y

vif $\rightarrow$ value for each X_i

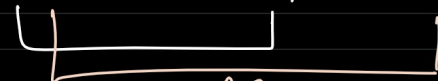
$X_{\text{lotArea}} \rightarrow \text{vif} \rightarrow \text{linear regression} \rightarrow y \rightarrow \text{lotArea}$

$X_{\text{YearBuilt}} \rightarrow \text{vif} \rightarrow y \rightarrow \text{YearBuilt}$
 $x_0, x_1 \rightarrow \text{lotArea, CrLiv Area}$

$X_{\text{CrLivArea}} \rightarrow \text{vif}$

$x_0, x_1 \rightarrow y \rightarrow \text{YearBuilt, CrLiv Area}$

$y \leftarrow x_0, x_1 \rightarrow y_{\text{pred}}, y_{\text{mean}}$



R^2

$\downarrow$

$$\frac{1}{1-R^2} = \text{vif}$$

$x_0 x_1 x_2 x_3 \rightarrow \text{all vif} \rightarrow$

$x_0 \rightarrow y$

$x_1 x_2 x_3 \rightarrow x$

$\rightarrow x_1 \rightarrow y$

$x_0 x_2 x_3 \rightarrow x$

Handling imbalanced data :

- 1) RU (Random Under Sampling)
- 2) RO (Random Over Sampling)
- 3) SMOTE (Synthetic Minority Over-Sampling Technique)
- 4) ADASYN (Adaptive Synthetic Sampling)

(1) RUS → balance data by randomly deleting rows from the majority class

(1a) A → 10 rows

B → 30 rows

C → 70 rows

D → 12 rows

↓

RUS

→ A → 10 rows

B → randomly delete → 20 rows → final data → 10 rows

C → 10 rows

D → 10 rows

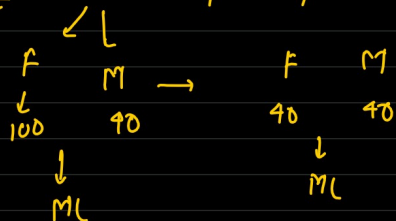
A → 1 milli
B → 1.2 mi
C → 1.3
D → 1.1

1 million

ID feature_x₁ feature_x₂ feature_y

-	-	-	1
-	-	-	0
-	-	-	1
-	-	-	2
-	-	-	3
-	-	-	4

students : class → p.s → get unbiased stats about teacher preferences(A)



100 → A

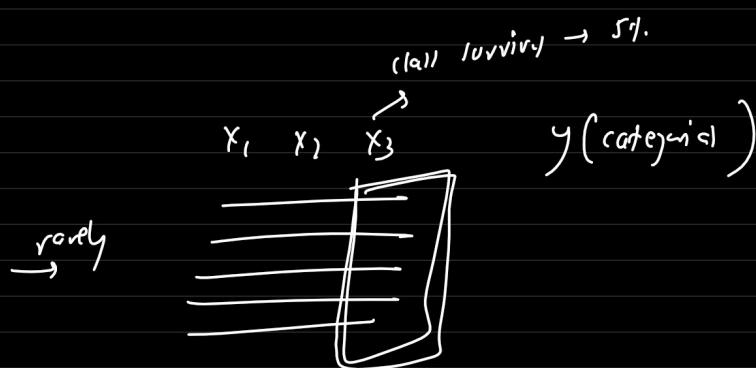
98 → B → RUS

102 → C

(2) ROS $\rightarrow$ balance data by random duplicating existing rows from the minority class

$m \rightarrow 40 \rightarrow 120$ (random duplication) $\rightarrow 60$
 $f \rightarrow 60 \rightarrow 60$

$max \rightarrow 60$
 $min \rightarrow 40 \rightarrow max-min$ (random duplication)



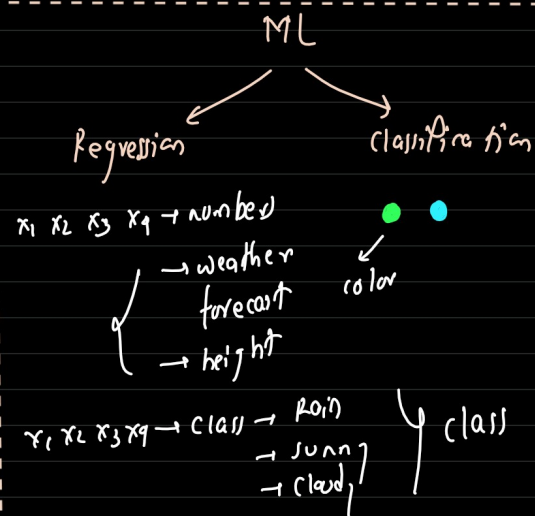
2000 row A
 1800 row $\rightarrow$ B $\rightarrow$ fire

2K A
 700 B $\rightarrow$

data imbalance $\rightarrow$ ML

Test
 data $\rightarrow$ A $\rightarrow$ B
 $\rightarrow$ A $\rightarrow$ A
 $\rightarrow$ B $\rightarrow$ B

A $\rightarrow 100\%$
 B $\rightarrow 50\%$
 $\rightarrow$ C $\rightarrow 10\%$
 D $\rightarrow 70\%$



$x_1, x_2, x_3 \rightarrow$ (cat) (A) $\rightarrow$ 95 row
 $x_1, x_2, x_3 \rightarrow$ (cat) (B) $\rightarrow$ 5 row

$x_1, x_2, x_3 \rightarrow$ A

95 row A $\rightarrow$ ML $\rightarrow$ (95%)
 5 row B

del (x_1, x_2) return A $\rightarrow 95\%$

x_1	x_2	...	x_{50}	y
				1
				1
				1
				2
				(+2)
x_1	...		x_{50}	

(3) SMOTE

(creates) entirely new synthetic minority rows

ID	x_1	x_2	$y(\text{class})$	
1	1	2	1	$y(1) \rightarrow 3$
2	2	4	1	$y(0) \rightarrow 2$
3	1.5	3.0	1	
4	8.0	9.0	0	
5	8.5	9.5	0	
6	8.2	9.2	0	

minority $\rightarrow$ class(0)
 $\rightarrow$ synthetic rows to create

$\rightarrow$ select base data $\rightarrow$ minimum in minority $\rightarrow$ (8.0, 9.0) Id. 4.

$\rightarrow$ nearest minority data $\rightarrow$ data(minority) $\rightarrow$ base data

Eucclidean distance: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

(8.5, 9.5) W.S.

$$\sqrt{(8.5 - 8)^2 + (9.5 - 9)^2} = \sqrt{(0.5)^2 + (0.5)^2} = \sqrt{0.25 + 0.25} = \sqrt{0.5} = 0.7$$

Example:

minority $\rightarrow$ A
B
C
D
E
F

step 1 select minimim point $\rightarrow$ base data (x) (A)

step 2 find closest data $\rightarrow$ X $\rightarrow$ B
X $\rightarrow$ C $\rightarrow$ euclidean
X $\rightarrow$ D dist
X $\rightarrow$ E
X $\rightarrow$ F

A - F closest point from Base

step 3 calculate vector difference

A $\rightarrow$ 8, 9

$$\Delta x = 8.5 - 8 = 0.5$$

B $\rightarrow$ 8.5, 9.5

$$\Delta y = 9.5 - 9 = 0.5$$

step 4: select any random number (λ) between 0 & 1

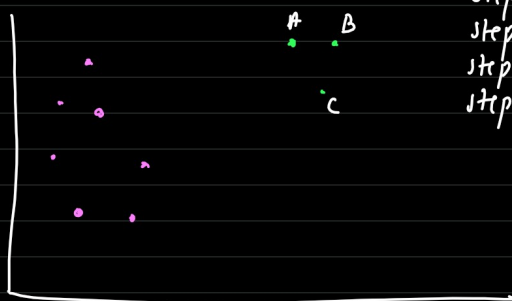
$$\lambda = 0.4$$

step 5: multiply the difference by λ & add to the base point

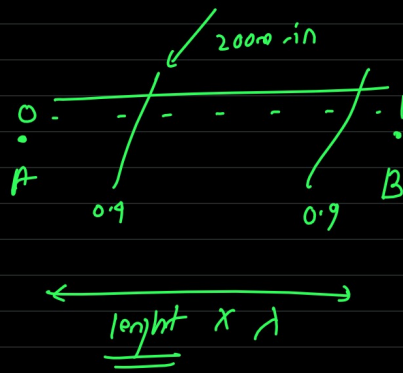
Base-point $\rightarrow$ (8, 9)

$$\text{New } x = x + (\Delta x \times \lambda) \rightarrow 8 + (0.5 \times 0.4) \rightarrow 8.2$$

$$\text{New } y = y + (\Delta y \times \lambda) \rightarrow 9 + (0.5 \times 0.4) \rightarrow 9.2$$



Step 1: Given
Step 2: Base $\rightarrow$ A
Step 3: A - B
Step 4:



4. ADASYN

- works like SMOTE but smart.
- looks at all minority points & checks how many majority neighbours surround them.
- It creates more synthetic points next to minority points that are trapped near majority data.



A → $k=3$

A → - - - → Danger

D → - - -

A
B → Danger $\frac{maj}{N}$
C
D

str → encode → ONE

imbalanced → number → sampling

↓
X

↓
categorical



→

RVS → majority →
minority →

sampling

$$0.5 = \frac{N_{\text{minority}}}{N_{\text{majority}}} = \frac{500}{1000}$$

$$1 = \frac{1000}{1000}$$